

1. Administrative & Study Identification

Full Study Title: A Study to Assess the Safety, Pharmacokinetics, and Activity of R07790121 in Participants With Advanced MASH Liver Fibrosis **Short Title/Acronym:** R07790121 MASH Fibrosis Study **Protocol Number:** CC45687 **NCT Number:** NCT06903065 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** R07790121 (anti-TL1A monoclonal antibody; also known as PF-06480605 / RVT-3101 / RG6631) **Phase of Development:** Phase 1 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, pharmacokinetics (PK), pharmacodynamics (PD), immunogenicity, and clinical activity of R07790121 in participants with advanced metabolic dysfunction-associated steatohepatitis (MASH) fibrosis. **Secondary Objectives:** To evaluate the effect of TL1A inhibition on liver fibrosis markers, MASH resolution, and systemic inflammatory biomarkers.

2.2 Background & Rationale To address the significant unmet medical need for patients with advanced metabolic dysfunction-associated steatohepatitis (MASH) complicated by liver fibrosis. R07790121 targets Tumor Necrosis Factor-like cytokine 1A (TL1A), a protein deeply involved in driving inflammation, tissue thickening, and fibrotic scarring. This study will evaluate if addressing the underlying inflammatory pathways can safely and effectively reduce liver scarring in MASH patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm / Dose-exploration **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 50 participants **Number of Sites:** Multicenter (US and Rest of World [ROW]) **Estimated Study Start:** 14-Apr-2025 **Estimated Primary Completion:** 30-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Body mass index (BMI) within the range of ≥ 25 and ≤ 45 kg/m². 2. MASH with fibrosis score of

F3 or F4 confirmed by transient elastography measurement ≥ 12.0 kPa and ≤ 25.0 kPa. 3. Agreement to adhere to strict contraception requirements.

4.2 Exclusion Criteria 1. Weight gain or loss $>5\%$ in the 3 months prior to baseline or $>10\%$ in the 6 months prior to baseline; or history of bariatric surgery within 1 year prior to baseline. 2. Current signs or prior history of decompensated liver disease, complications/clinical evidence of portal hypertension, or current/prior history of hepatocellular carcinoma (HCC). 3. Concomitant Type 1 diabetes, or Type 2 diabetes with HbA1c $>10\%$; active tuberculosis; or initiation of new weight loss, antidiabetic, lipid-modifying, or anti-depressant medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants will receive R07790121 via intravenous (IV) infusion on Day 1, and Weeks 2, 6, and 10. Participants will then transition to R07790121 subcutaneous (SC) injections every four weeks (Q4W) from Week 14 up to and including Week 50. **Route of Administration:** Intravenous (IV) infusion followed by Subcutaneous (SC) injection. **Duration of Treatment:** 50 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background medications for metabolic conditions (if initiated prior to study cutoffs). **Prohibited:** Initiation of new medications from antidiabetic, weight loss, lipid-modifying, or anti-depressant drug classes during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Transient Elastography, Baseline Labs
Baseline	Day 1	First IV Infusion, Baseline PK/PD/Immunogenicity Labs
Interim	Weeks 2, 6, 10, then Q4W	IV Infusions (Wks 2/6/10) followed by Q4W SC Injections, Safety Labs, PK/PD tracking
End of Study	Week 50+	Final Dosing, Efficacy/Fibrosis Assessment, Follow-up Labs, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs), alongside core PK and PD parameters up to study completion. **Measurement Method:** Clinical laboratory tests, vital signs, physical examinations, and centralized plasma concentration assays.

7.2 Secondary & Exploratory Endpoints * **Immunogenicity:** Incidence of anti-drug antibodies (ADAs) against R07790121. * **Clinical Activity:** Change from baseline in liver fibrosis severity and MASH resolution assessed via transient elastography/centralized imaging and/or histology.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via HCP observation and patient reporting at each onsite clinical visit. **Serious Adverse Events (SAEs):** Reported by the investigator within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring committee managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving at least one dose) and PK/PD-evaluable Set. **Sample Size Justification:** Target sample of $N=50$ is deemed adequate to capture preliminary Phase 1 safety, tolerability signals, and construct PK/PD modeling curves. **Handling of Missing Data:** Standard mixed-effects modeling for repeated measures; appropriate imputation strategies for PK concentrations below the limit of quantification.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of source data, focusing on safety reporting and dose administration fidelity. **Audit Trail:** Robust eCRF locking and query resolution via standard data clarification forms; centralized review of liver histology and elastography scans.

11. Study Identifiers & Governance

Primary ID: CC45687 **Registry Number:** NCT06903065 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study to Assess the Safety, Pharmacokinetics, and Activity of R07790121 in Participants With Advanced MASH Liver Fibrosis **Official Regulatory Title:** A Study to Assess the Safety, Pharmacokinetics, and Activity of R07790121 in Participants With Advanced MASH Liver Fibrosis

12. Clinical Framework (MASH Specific)

Primary Condition: Metabolic Dysfunction-Associated Steatohepatitis (MASH) Liver Fibrosis **Diagnosis Threshold:** Fibrosis score F3 or F4 (Transient elastography ≥ 12.0 kPa and ≤ 25.0 kPa) **Medical Methodology:** Anti-TL1A monoclonal antibody therapy **Optimization Target:** Reduction in inflammatory signaling, MASH resolution, and reversal of liver scarring

13. Study Procedures & Methodology

Management Pathway: Targeted biologic intervention and strict monitoring for advanced MASH patients **Education Protocol (HCP):** Training on protocol-specific IV infusion and SC injection sequencing **Education Protocol (Patient):** Contraception counseling, weight maintenance, and reporting of AEs **Quality Audit Mechanism:** Centralized evaluation of baseline and follow-up liver histology/elastography

14. Observation & Follow-Up Schedule

Total Duration: ~62 weeks (Screening + 50-week treatment duration + safety follow-up) **Onsite Visit Cadence:** Day 1, Weeks 2, 6, 10 (IV dosing) followed by Every 4 weeks (Q4W) for SC dosing **Remote Monitoring Frequency:** As required per protocol between monthly clinic visits **Checklist Review Interval:** At every dosing visit (Q4W)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						